Roll No. 23E1011 **BE I EXAMINATION January 2024** **2024276**

(Branch: ETC B and E& I)

APPLIED PHYSICS (2APRC2)

Duration: 3 hrs. **Max.Marks:60**

NOTE: All questions are compulsory. Out of two parts in each question attempt any one part.

Q1 I (a) Explain the principle and construction of the Michelson's interferometer. Explain the formation of the different type of the fringes. **8**

(b) Explain the diffraction from the circular aperture. In a plane transmission grating, the angle of diffraction for second order principal maxima for wavelength 5000Å is 30^{o}. Calculate the number of lines in the grating. **4**

OR

II(a) What are Fresnel's half period zones? Prove that the area of a half period zone on a plane wavefront is independent of the order of the zone. **8**

(b) Explain why Newton's fringes are call fringes of equal thickness. In Newton's rings experiment the diameter of the 4^{th} and 25^{th} dark rings are 0.3cm and 0.8 cm, respectively. Find the wavelength of light, if radius of curvature of lens is 100cm. **4**

Q2 I (a) Explain working, construction and energy level diagram of CO_2 laser. Also explain the origin of the energy levels in this laser. **8**

(b) What is Malus law? Find the thickness of the quarter wave plate for wavelength 5890Å if the birefringes of material is 0.031. **4**

OR

II(a) Derive the expression for the production of the circularly and elliptically polarized light. **6**

(b) Describe the construction of optical fiber and its type depends on material, refractive index and mode. For a step index fiber with a core of the refractive index 1.52 and the cladding is doped to give a fractional index difference 0.0006. Then find cladding refractive index. **6**

Q3 I(a) What is lattice plane? Derive an expression for inter planer distance. **6**

(b) Write notes on (i) Band theory and (ii) Transistor. **6**

OR

II(a) Explain the formation of depletion region in a pn diode and effect of forward and reverse biased voltage on it. **06**

(b) Write notes on (i) Miller Indices and (ii) Fermi energy level. **06**

Q4	I(a)	State and explain the charge continuity equation for the time varying current distribution and show that it also satisfied the steady state condition.	06
	(b)	Derive the wave equation of electric and magnetic field for free space and show that the velocity of plane electromagnetic wave in free space is $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.	6

OR

	II(a)	Explain the inconsistency in Amper's law and derive the expression for the displacement current density.	8
	(b)	Using Maxwell's equation n show that the magnetic mono poles does not exist.	4
Q5	I (a)	What is the Planck's Radiation law and its postulates? Drive the expression for Planck's radiation law?	8
	(b)	What is phase velocity and group velocity? Derive the relation between them.	4

OR

	II(a)	Derive the time independent wave equation for 3 dimension and also write down for the free partial.	6
	(b)	What is wave function? Describe the Davisson and Germer experiment and its importance. Find the energy of the neutron in eV whose de-Broglie wavelength is 1 Å? [m=1.674x10^{-27}kg]	6